THE HIGHER DERIVED LOWER SHRIEK OF AN ABELIAN SHEAF ALONG A MORPHISM OF FIBRED CATEGORIES IS COMPUTED BY THE HOMOLOGY OF THE INDEX CATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the description of the index category and the presheaf, the amelioration of the morphism with the exactness of the first factor, the reduction to the fibred case, and the final identification of the index categories.

Editorial note. The label lemma-compute-left-derived-g-shriek names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing situation and to the index categories introduced in the proof; the seven cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Situation of a morphism of fibred categories and its lower shriek g_!. Same chapter, the standing situation of the morphism of fibred categories and the adjoint pair (g_!, g^*). Supplies the object whose derived functors are computed and the index category appearing in the formula.
- Amelioration of a morphism of fibred categories. Categories, the lemma factoring u through an ameliorated morphism u" with an exact u'. Supplies the factorisation u = u" o u' used to split the computation into an exact part and the general case.
- Exactness of g'_! and the computation lemma for the fibred case. Modules on Sites, the exactness of the lower shriek for the ameliorated morphism, and Same chapter, the computation of L_n pi_! for a fibred category. Supplies Lg_! = Lg"_! after applying the exact g'_!, to which the known computation lemma applies.

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Source reference. Original lemma 08NF: sources/stacks-chow/tags.txt, source lines 10988–10988. Prerequisite pointer invoked at proof line(s) 10709; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PJ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 10615–10615. Prerequisite pointer invoked at proof line(s) 10721; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08NH: sources/stacks-chow/tags.txt, source lines 10990–10990. Prerequisite pointer invoked at proof line(s) 10718; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00XR: sources/stacks-chow/tags.txt, source lines 1229–1229. Prerequisite pointer invoked at proof line(s) 10731; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08P3: sources/stacks-chow/tags.txt, source lines 11011–11011. Prerequisite pointer invoked at proof line(s) 10712; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PA: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9767–9767. Prerequisite pointer invoked at proof line(s) 10680; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06NW: sources/stacks-chow/tags.txt, source lines 8550–8550. Prerequisite pointer invoked at proof line(s) 10729; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assumptions and notation as in Situation 08PA. For an abelian sheaf $\mathcal{F}'$ on $\mathcal{C}'$ the sheaf $L_n g_!(\mathcal{F}')$ is the sheaf associated to the presheaf*

$$U \longmapsto H_n(\mathcal{I}_U, \mathcal{F}'_U)$$

For notation and restriction maps see proof.

Proof. Say $p(U) = V$. The category $\mathcal{I}_U$ is the category of pairs (U', φ) where $\varphi : U \to u(U')$ is a morphism of $\mathcal{C}$ with $p(\varphi) = \mathrm{id}_V$, i.e., φ is a morphism of the fibre category $\mathcal{C}_V$. Morphisms $(U'_1, \varphi_1) \to (U'_2, \varphi_2)$ are given by morphisms $a : U'_1 \to U'_2$ of the fibre category $\mathcal{C}'_V$ such that $\varphi_2 = u(a) \circ \varphi_1$. The presheaf $\mathcal{F}'_U$ sends (U', φ) to $\mathcal{F}'(U')$. We will construct the restriction mappings below.

Choose a factorization

$$\mathcal{C}' \underset{w}{\overset{u'}{\rightleftarrows}} \mathcal{C}'' \xrightarrow{u''} \mathcal{C}$$

of u as in Categories, Lemma 08NF. Then $g_! = g''_! \circ g'_!$ and similarly for derived functors. On the other hand, the functor $g'_!$ is exact, see Modules on Sites, Lemma 08P3. Thus we get $Lg_!(\mathcal{F}') = Lg''_!(\mathcal{F}'')$ where $\mathcal{F}'' = g'_!\mathcal{F}'$. Note that $\mathcal{F}'' = h^{-1}\mathcal{F}'$ where $h : Sh(\mathcal{C}'') \to Sh(\mathcal{C}')$ is the morphism of topoi associated to w, see Sites, Lemma 08NH. The functor u'' turns $\mathcal{C}''$ into a fibred category over $\mathcal{C}$, hence Lemma

08PJ applies to the computation of $L_n g''_!$. The result follows as the construction of $\mathcal{C}''$ in the proof of Categories, Lemma 08NF shows that the fibre category $\mathcal{C}''_U$ is equal to $\mathcal{I}_U$. Moreover, $h^{-1}\mathcal{F}'|_{\mathcal{C}''_U}$ is given by the rule described above (as w is continuous and cocontinuous by Stacks, Lemma 06NW so we may apply Sites, Lemma 00XR). $\square$